

Bellabeat Case Study

Smart Device Usage Analysis

Data Analytics Project

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Tools Used: R (tidyverse, ggplot2, dplyr)

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Ask

Business Task

The business task is to analyze data from non-Bellabeat smart devices to identify trends in how consumers track activity, sleep, and other health behaviours. These insights will be used to inform the Bellabeat team and provide high level recommendations for marketing strategies and product positioning.

Stakeholders

- Urška Sršen - Co-founder and Chief Creative Officer, Bellabeat.
- Sando Mur - Co-founder and mathematician; member of the Bellabeat executive team.
- Bellabeat Marketing Analytics Team - Internal team responsible for analyzing and applying insights to marketing campaigns.

Key Guiding Questions

- What are trends in smart device usage?
- How can these trends apply to Bellabeat customers?
- How can they inform marketing strategy?

Prepare

Primary Dataset: [Fitbit Fitness Tracker Dataset](#) - Mobius

Data Sources

- The dataset used in this analysis is the Fitbit Fitness Tracker Data obtained from Kaggle. The dataset contains daily activity data from 33 Fitbit users, including steps, calories, and activity levels.

Data Organization

The data is organized into rows and columns in long format. Each row represents a singular user's daily activity record. Each column represents the variables such as ID, total distance, calories burned, total steps, and the level of activity intensity.

The dataset mostly consists of numerical variables with one character variable representing dates. This allows for analysis of the daily activity trends of users.

Credibility and Bias

- 33 users - small sample
- The data does not pass the credibility test. The data claims there are 30 users but upon further analysis the data consists of 33 participants. There is also very little information about the demographics of the participants. The data does not present the broader population, a larger size would be more useful for smart device users, this could introduce bias into the analysis.
- This data was also collected in 2016 making it slightly outdated.
- The data is likely self-reported and collected through the participants' smart devices. This could introduce bias and inaccuracies.
- The data is publicly available and offers real user data making it suitable for exploratory purposes.

Licensing, Privacy, Security, Accessibility

- CC0: Public Domain - Free to use, the data is anonymous; there are no names, just IDs to show participants. The dataset is stored locally and will be shared securely through the project repository, ensuring accessibility while maintaining data privacy and security standards.

Data Integrity

- Data checks were performed using summary statistics and structural analysis in R on three different datasets, looking at Daily Activity, Daily Calories, and Daily Steps. The data was reviewed for missing values and unusual values, along with inconsistencies.
- No major issues were identified to effect analysis, however in the next stage, further cleaning will be performed to address any further inconsistencies.

Relevance

- The data is relevant to the business task as it provides important insights on users habits and trends, it looks at how users engage with their smart devices to track activity and health behaviours. These patterns can be used to identify trends in user behaviour, which will help inform marketing strategies and product positioning for Bellabeat.

Potential Problems

- There are several limitations, some of which were mentioned above that exist within this dataset.
- The sample size is small and this analysis would benefit from a larger one when making informed decisions.
- The lack of demographic information also limits the ability to segment users and apply findings to specific target groups.

Secondary Dataset: [Fitness Tracker Dataset](#) - AJ

Data Sources

- This dataset is the Fitness Tracker Data obtained from Kaggle. The dataset contains daily activity data from various users including similar information from our first dataset - such as daily activity, steps, and calories. This secondary dataset was included in this project for exploratory comparison purposes. While summary statistics were calculated, the primary focus of the analysis remained on the Fitbit Fitness Tracker Data due to its completeness and relevance to the research questions.

Data Organization

The data is organized into rows and columns in long format. Each row represents a singular user's daily activity record. Then each column represents the variables such as steps, calories burned, distance, and active minutes.

The dataset contains positive integers, characters, and numeric variables.

Credibility and Bias

- The dataset contains 999 unique users, indicating a relatively large sample size compared to the Fitbit dataset. This increases the reliability of identifying general activity trends. However, there is limited demographic information available about the participants, such as age and gender. This makes it difficult to determine how representative the dataset is of the broader population.
- The data is likely self-reported and collected through the participants' smart devices. This could introduce bias and inaccuracies.
- Data was collected in 2023 making it relatively current and relevant for our analysis and comparison
- Overall, while the dataset is useful for exploratory analysis and identifying trends, potential bias and lack of demographic detail should be considered when interpreting results.

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Data Integrity

- Data checks were performed using summary statistics and structural analysis in R.
- The data was reviewed for missing values and unusual values, along with inconsistencies.
- No major issues were identified to effect analysis, however in the next stage, further cleaning will be performed to address any further inconsistencies.

Relevance

- The dataset is relevant to the business task as it provides insight into how users engage with their smart devices to track activity, sleep, and overall health behaviours. These insights can be used to identify trends in user behaviour, which

can help inform data-driven marketing strategies and product positioning for Bellabeat.

Potential Problems

- Several potential limitations exist within the dataset. Although the sample size is relatively large, there is limited demographic information available, which restricts the ability to generalize findings to specific populations.
- Additionally, some variables, such as mood, may be self-reported and therefore subject to bias or inconsistency.
- Differences in how data is recorded across devices or users may introduce inconsistencies that should be considered when interpreting results.

Process

Tools Used

- The data was processed and cleaned using R, primarily utilizing the tidyverse package for efficient data manipulation and transformation. R was selected for its ability to handle large datasets, perform data cleaning, and support reproducible analysis..
- The analysis incorporates multiple datasets, including Fitbit data files such as:
 - dailyActivity_merged
 - dailyCalories_merged
 - dailySteps_merged
- As well as a secondary dataset (Fitness Tracker Dataset by AJ) to supplement and strengthen the analysis.

Data Integrity Checks

- Initial checks were performed to ensure the dataset was accurate and usable:
 - Reviewed structure and data types using glimpse() and str()
 - Generated summary statistics using summary()
 - Checked for missing values using sum(is.na())
 - Verified column names and consistency using colnames()
- These checks confirmed that the dataset was well-structured, with no major issues preventing analysis.

Data Cleaning Steps

1. I looked for missing values: All datasets had no missing values.
2. Next I checked for duplicates: There were no duplicates in any of the datasets.
3. Converted date columns into proper date format.
4. Created a new variable (day of the week) to support trend analysis.
5. For the second_dataset I checked for duplicate records, as well as cleaned the data using R to ensure that only unique observations were included in the analysis.
6. Reviewed summary statistics to identify potential outliers.

Data Transformation

Data was transformed to improve usability for analysis. Date columns were converted into proper date format, and a new variable representing the day of the week was created to support weekly trend analysis.

These transformations ensured the dataset was structured appropriately for aggregation and analysis in later stages.

Data Cleaning Verification

After cleaning I reviewed the data to ensure accuracy and consistency. Summary statistics were also rechecked to ensure the variables were in their expected ranges and to make sure no unexpected changes occurred during the cleaning process.

The structure and data types of the datasets were reviewed to ensure the variables were correctly formatted and ready to use for analysis.

Documentation of Cleaning Process

All data cleaning and transformation was done in R scripts and documented within the R script document. This ensures that the process is reproducible and can be reviewed or modified if needed.

Analyze

Data Aggregation

Data from the Fitbit dataset was aggregated to identify patterns in users activity across the week. The average daily steps were approximately **7,638 steps**, while the average calories burned were approximately **2,304 calories**. These values provide us with an overview of user activity levels.

To add comparison, the Fitness Tracker dataset was also summarized. This dataset showed higher average daily activity levels, with approximately **15,006 steps** and **2,750 calories burned per day**.

These aggregated values allow us to establish what baseline activity levels may look like and allow for comparison of user behaviour across different datasets.

Data Organization and Formatting

- The datasets were organized as data frames in R, where each row represents an individual observation and each column represents variables such as steps, calories, and date.
- The Fitbit dataset and the secondary fitness tracker dataset were imported separately to maintain clarity during analysis. Column names and variable structures were reviewed to ensure consistency across datasets.
- This organization enabled efficient grouping, summarization, and comparison of user activity patterns across time.

Descriptive Analysis and Calculations

Descriptive statistics were used to summarize user activity and energy expenditure within the dataset.

The **average number of daily steps was approximately 7,638**, with a **median of 7,406 steps**. The **range of steps was wide, from 0 to 36,019**, indicating significant variation in activity levels among users. The **standard deviation of 5,087** further supports the presence of substantial variability in daily step counts.

Calorie expenditure showed an **average of approximately 2,304 calories per day**, with a **median of 2,134 calories**. **Values ranged from 0 to 4,900 calories**, suggesting differences in daily energy output across users. The **standard deviation of 718** indicates moderate variability in calorie burn compared to step counts.

Overall, the results suggest that while average activity levels are moderate, there is a wide spread in user behaviour, with some individuals being significantly more active than others.

Trends and Relationships

Analysis of daily activity patterns revealed trends in user behaviour across the week. Average daily step counts varied by day, with the highest levels observed on Saturday (8,153 steps) and Tuesday (8,125 steps). The lowest average step count occurred on Sunday (6,933 steps). This indicates that user activity is not consistent throughout the week, with some days showing higher levels of engagement than others.

A similar pattern was observed in calorie expenditure. The highest average calories burned occurred on Tuesday (2,356 calories) and Saturday (2,355 calories), while the lowest values were recorded on Thursday (2,200 calories) and Sunday (2,263 calories). This suggests that calorie burn follows a pattern similar to step count across the week. The relationship between steps and calories indicates a positive correlation, where increased physical activity is associated with higher energy expenditure.

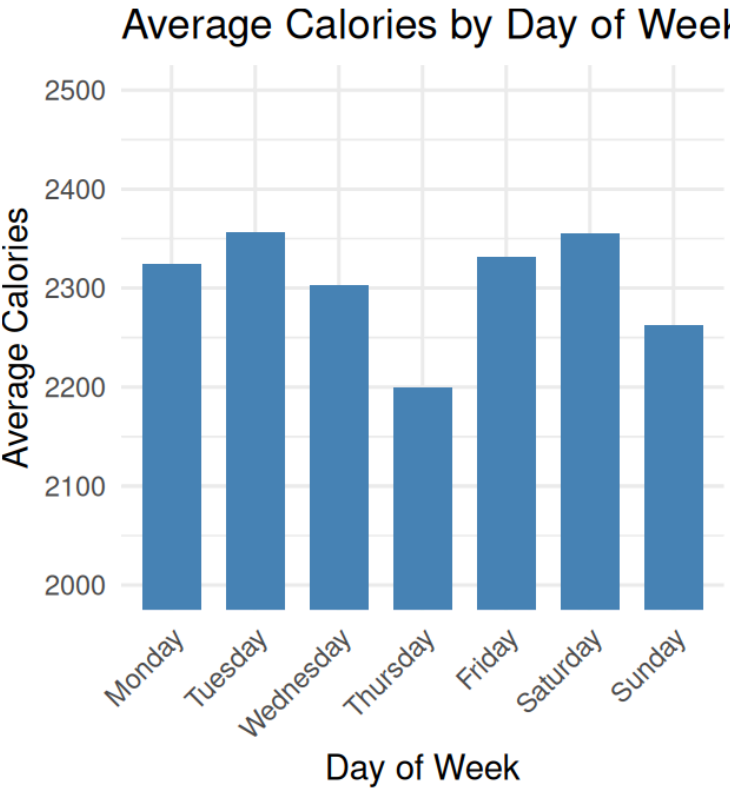
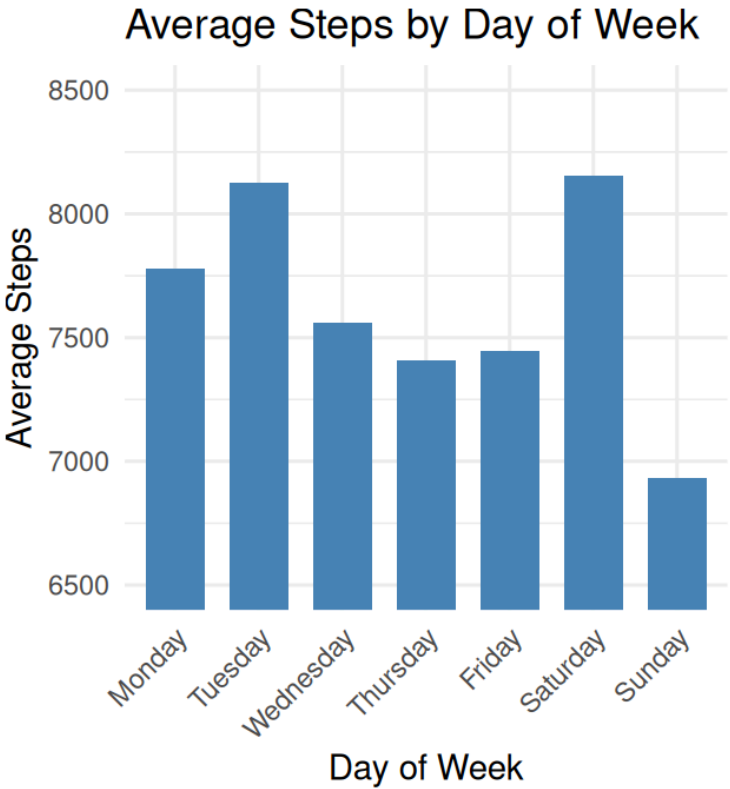
In terms of consistency, there is a high level of variation in daily step counts, while calorie expenditure shows less variability. This suggests that users are inconsistent in their activity levels, although overall energy output remains relatively more stable.

These patterns suggest opportunities to encourage increased activity on lower-activity days, such as Sundays.

Key Insights Related to Business Questions

- The analysis of smart device usage data provides several key insights that can inform Bellabeat's marketing strategy:
 - First, user activity levels vary significantly across the week, with lower activity observed on certain days such as Sundays. This suggests that users may be less engaged in physical activity during these periods. Bellabeat can leverage this insight by introducing targeted notifications, reminders, or challenges to encourage increased activity on lower-engagement days.
 - Second, a clear positive relationship was observed between step count and calorie expenditure, indicating that increased physical activity directly contributes to higher energy output. Bellabeat can use this insight to emphasize the connection between movement and health outcomes in their marketing, reinforcing the value of tracking activity through their products.

Average Steps and Calories by Day of Week



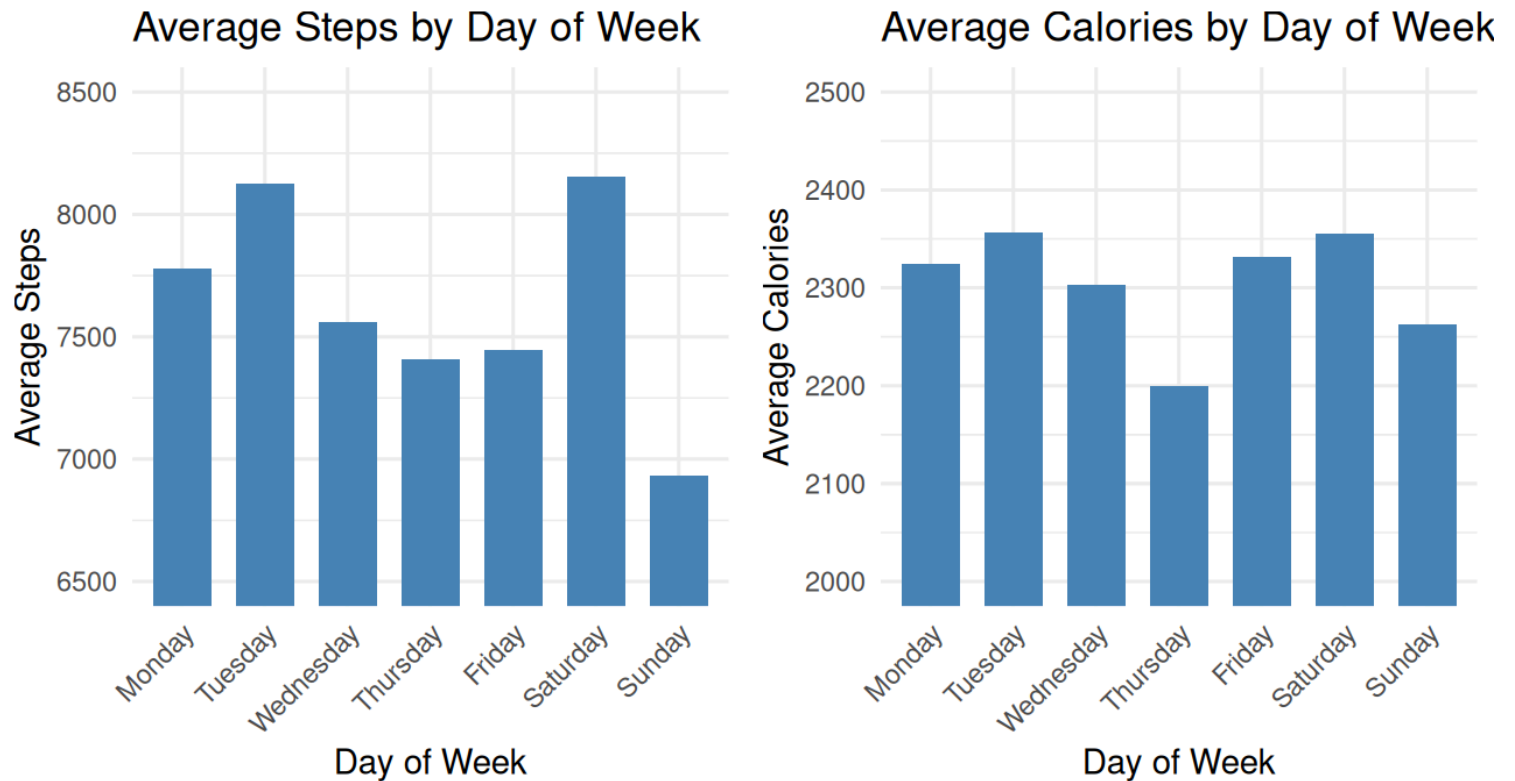
Steps and calorie expenditure were analyzed across the week using aggregated averages. Both metrics were visualized side-by-side to allow for comparison across days of the week.

The data indicates that activity levels vary by day, with higher values observed on Tuesday and Saturday and lower values on Sunday. This pattern is consistent across both steps and calorie expenditure, suggesting a relationship between physical activity and energy output.

Share

The following visualizations summarize key patterns in user activity and energy expenditure across the week.

Average Steps and Calories by Day of Week



These visuals highlight that user activity varies by day, with lower engagement on Sundays and higher engagement mid-week and on weekends.

This suggests an opportunity for Bellabeat to implement targeted engagement strategies, such as reminders or challenges, to encourage more consistent activity.

Overall, these insights support the development of more personalized and data-driven user engagement strategies.

Act

Final Recommendations

I identified clear patterns in smart device usage and physical activity behaviour across the week. User engagement levels varied depending on the day, with higher activity observed on days such as Tuesday and Saturday and lower activity on Sundays.

These findings also demonstrated a consistent relationship between physical activity and calorie expenditure, suggesting that users who engage in more movement also experience higher energy output. These insights provide Bellabeat with opportunities to strengthen user engagement through more personalized and data driven strategies.

3 Top Recommendations

1. Targeted Engagement Reminders: Bellabeat could introduce targeted reminders, notifications, or weekly activity challenges on lower-engagement days such as Sundays to encourage more consistent physical activity among users.
2. Expand Personalized Wellness Insights: The variability observed in user activity levels suggests an opportunity for Bellabeat to provide more personalized fitness recommendations, adaptive goals, and tailored wellness insights based on individual behaviour patterns.
3. Strengthen Market Around Activity Tracking Benefits: Bellabeat can use the demonstrated relationship between movement and calorie expenditure in marketing campaigns to reinforce the health benefits of consistent activity tracking through Bellabeat products such as Leaf and the Bellabeat app.

Future Considerations

For future analysis it would be recommended to incorporate larger and more diverse datasets to better understand differences across demographics, lifestyles, and geographic regions. Additional data related to stress, hydration, or sleep patterns may also provide deeper insight into overall wellness behaviours.